Rethinking Transformers for Semantic Segmentation of Remote Sensing Images

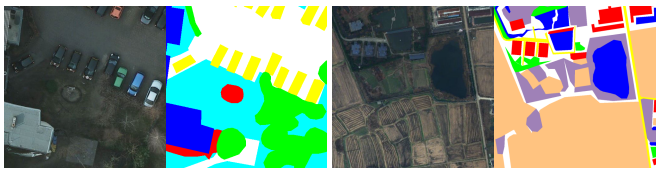
Yuheng Liu, Yifan Zhang, *Member, IEEE*, Ye Wang, and Shaohui Mei, *Senior Member, IEEE*

***Abstract*— Transformer has been widely applied in image processing tasks as a substitute for convolutional neural networks (CNNs) for feature extraction due to its superiority in global context modeling and flexibility in model generalization. However, the existing transformer-based methods for semantic segmentation of remote sensing (RS) images are still with several limitations, which can be summarized into two main aspects: 1) the transformer encoder is generally combined with CNN-based decoder, leading to inconsistency in feature representations; and 2) the strategies for global and local context information utilization are not sufficiently effective. Therefore, in this article, a global-local transformer segmentor (GLOTS) framework is proposed for the semantic segmentation of RS images to acquire consistent feature representations by adopting transformers for both encoding and decoding, in which a masked image modeling (MIM) pretrained transformer encoder is adopted to learn semantic-rich representations of input images and a multiscale global-local transformer decoder is designed to fully exploit the global and local features. Specifically, the transformer decoder uses a feature separation-aggregation module (FSAM) to utilize the feature adequately at different scales and adopts a global-local attention module (GLAM) containing global attention block (GAB) and local attention block (LAB) to capture the global and local context information, respectively. Furthermore, a learnable progressive upsampling strategy (LPUS) is proposed to restore the resolution progressively, which can flexibly recover the fine-grained details in the upsampling process. The experiment results on the three benchmark RS datasets demonstrate that the proposed GLOTS is capable of achieving better performance with some state-of-the-art methods, and the superiority of the proposed framework is also verified by ablation studies. The code will be available at https://github.com/lyhnsn/GLOTS.**

***Index Terms*— Encoder–decoder structure, global-local transformer, remote sensing (RS), semantic segmentation.**

I. Introduction

IT IS increasingly essential to extract useful information from remote sensing (RS) images in many applications, such as urban planning [1], [2], [3], disaster management [4], [5], [6], environmental monitoring [7], [8], [9], and other scenes [10], [11], [12], with the rapid development of RS technology providing a large amount of high spatial resolution (HSR) images. One of the most beneficial research directions is semantic segmentation of RS images, which can contribute to a more comprehensive understanding of the environment and land cover [13], [14], [15], [16], [17]. Semantic segmentation of RS images refers to assigning a semantic label to each pixel in the image, which is a challenging task due to the large spatial scale of RS images, the heterogeneous object samples, and the multiscale features, as shown in Fig. 1.



(a) (b)

Fig. 1. Examples of the complexity of RS images. (a) Image and its label from the ISPRS Potsdam dataset [40]. (b) Image and its label from the loveDA dataset [41].

As the most commonly used methods to address these issues, convolutional neural networks (CNNs) have been widely and successfully applied in both natural image segmentation [18], [19], [20], [21] and RS visual tasks [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32]. CNNs own strong inductive bias, including spatial locality and translation invariance, which are beneficial for image processing tasks. However, these also make CNNs unable to capture the global context of the input image and are hard to be built deeper and wider. In recent years, transformer [33] has been explored as a substitute for CNNs in image processing tasks [34] by regarding the input image as a sequence of patches and using self-attention to acquire global representations of the input image. Transformers have almost no inductive bias, which makes them more flexible in model generalization and more scalable in model depth and width. Though transformers are initially unstable to train and are more likely to overfit when trained on small datasets, these issues have been greatly alleviated by the recent efforts of introducing data structure priors to transformers [35], [36], [37], [38], [39].

One of the most successful trials in improving the architecture of transformers is the Swin transformer [39], which is a transformer with a hierarchical structure and a shift window mechanism for information interaction. Swin transformer has shown excellent performance in natural scene image processing tasks and has been widely adopted in RS image segmentation [42], [43], [44], [45], [46]. These works either focus on designing a CNN-based decoder by adopting

Manuscript received 14 April 2023; revised 13 June 2023 and 25 July 2023; accepted 2 August 2023. Date of publication 4 August 2023; date of current version 16 August 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62171381. *(Corresponding authors: Yifan Zhang; Shaohui Mei.)*

The authors are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: hnlyh@mail.nwpu.edu.cn; yifanzhang@nwpu.edu.cn; wy2017263322@mail.nwpu.edu.cn; meish@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3302024

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.

the Swin transformer as the encoder or combine the Swin transformer with CNNs for feature extraction, demonstrating that the Swin transformer is robust for image modeling. However, the transformer encoder and CNN decoder are not perfectly compatible, which may result in inconsistent feature representations. Furthermore, the Swin transformer may be unsuitable for global feature extraction of RS images, as the RS images are usually much larger than natural images, while the Swin transformer is incapable of capturing the global semantic context on a large scale.

Instead, in this article, a novel framework named global-local transformer segmentor (GLOTS) is proposed for the semantic segmentation of RS images, in which transformers are employed as both encoder and decoder for consistent feature representations. Specifically, GLOTS uses the original vision transformer (ViT) [34] as the encoder for global context modeling and adopts the masked image modeling (MIM) pretraining [47] method to produce more semantic features of images. Compared with introducing inductive bias of CNNs (e.g., Swin transformer), the MIM-pretrained transformer encoder does not modify the architecture of ViT but is more flexible for global semantic information acquisition of RS images. The delicately designed decoder consists of a feature separation-aggregation module (FSAM) for multiscale representations acquisition, a global-local attention module (GLAM) for global-local context capturing, and a learnable progressive upsampling strategy (LPUS) for progressive resolution restoration. To complement the lack of multiscale representations in ViT, GLOTS adopts an FSAM to separate the encoded features into multiple scales for multiscale modeling and produce extra aggregation details for segmentation performance enhancement. For adequate global-local context modeling, GLOTS flexibly uses global attention block (GAB), local attention block (LAB), and shifted attention block (SAB) for global-local feature acquisition and information interaction. In addition, an LPUS is used to restore the resolution of encoded features, which can achieve more competitive performance than the widely used multiscale fusion strategy, and is more flexible compared with the fixed bilinear interpolation. The main contribution of this article is summarized as follows.

1) A novel framework, named GLOTS, is proposed for semantic segmentation of RS images, in which transformers are employed in both encoder and decoder to acquire consistent feature representations. The proposed GLOTS is capable of adequately modeling semantic information of RS images with the MIM-pretrained transformer encoder, and the delicately designed decoder is presented to be more compatible with the transformer encoder than the existing and popular CNN-based decoders.
2) In the transformer decoder, an FSAM is specifically proposed for multiscale object information generation, which is lacking in the original ViT. The multiscale features are essential for the recognition of complex objects and detailed small samples. The FSAM is also designed to parse the aggregation features for the decoder, enhancing the hidden information.
3) A GLAM is also proposed in the decoder to flexibly utilize GAB, LAB, and SAB to acquire global and local features as well as interact with each other to fuse the semantic and appearance information of RS images adequately.

The rest of this article is organized as follows. Section II introduces the related works of semantic segmentation with CNNs, semantic segmentation with transformer encoders, and methods involving transformer decoders. Then the overall framework of the proposed GLOTS is presented in Section III, each component of which is concretely described. In Section IV, both comparison experiments and ablation studies are applied, and the experiment results are presented and analyzed. Finally, the conclusion is drawn in Section V.

II. Related Works

A. Semantic Segmentation With CNNs

Semantic segmentation of RS images with CNNs is generally developed from natural scene image segmentation, the most influential works of which are fully convolutional network (FCN) [18] and U-net [19]. FCN produces an end-to-end protocol for semantic segmentation, and U-net establishes a symmetric encoder–decoder structure for feature extraction and reconstruction. Following these two works, excellent methods are proposed to improve the performance of semantic segmentation, such as PSPNet [48], PSANet [49], UperNet [50], and DeepLabv3+ [51]. However, most of these methods are based on the highly effective ResNet [52]. To improve the capability of CNNs, HRNet [21] is proposed to maintain the high resolution of feature maps. On this basis, Wu et al. [53] proposed an optimized HRNet, which explores the applications of HRNet in semantic segmentation. These above-mentioned methods are widely used in RS image segmentation tasks and have achieved remarkable performance. In order to address the specific characteristics of RS images properly, some methods are specifically proposed for RS image segmentation. For example, Guo et al. [54] proposed a learnable gated convolutional network (LGCN) for flexible information fusion, and Boulila [55] proposed a two-level feature extraction strategy for refinement. HRNet-based methods are also developed for RS image segmentation, such as Zhang et al. [56] employed HRNet for high-resolution context aggregation, and Sun and Zheng [57] combining HRNet with PSPNet for multiband RS image segmentation.

B. Semantic Segmentation With Transformer Encoder

The first trial of semantic segmentation based on the transformer is made by segmentation transformer (SETR) [58], which regards semantic segmentation from a sequence-to-sequence perspective and explores three different decoder designs. Segmenter [59] and SegFormer [60] make further efforts to prove the feasibility of semantic segmentation using transformers and achieve improved performance. In addition, transformers are also delicately designed as encoders and combined with CNN decoders for semantic segmentation, such as the MIM-pretrained method, BeiT [47], and the CNN-liked

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.

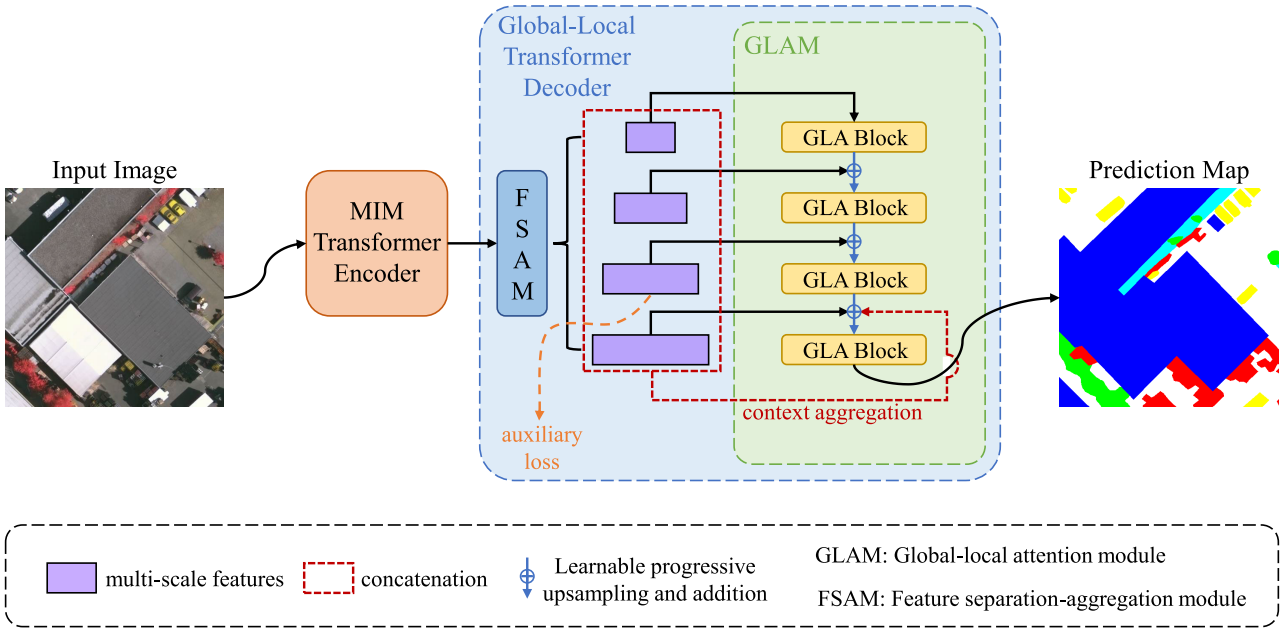


Fig. 2. Framework of GLOTS.

method, Swin transformer [39]. Methods for semantic segmentation of RS images with transformers are mainly built on the Swin transformer due to its outstanding capability in image modeling. For example, Zhang et al. [43] proved the effectiveness of the Swin transformer encoder, Xu et al. [45] used the tiny version of Swin transformer for efficient RS image segmentation, Wang et al. [44] proposd a densely connected module for feature fusion. He et al. [42] and Gao et al. [46] combined the Swin transformer with ResNet to build a hybrid network for parallel feature extraction. All the above methods are based on transformer encoders, combined with specifically designed CNN decoders. However, performance improvement is highly expected by exploring the potentiality of transformer-based decoders.

C. Transformer Decoder

A transformer is first proposed to be composed of an encoder and a decoder in natural language processing (NLP) task [33] to achieve autoregressive language modeling. When applied to image recognition tasks, the transformer encoder is pretrained on large-scale datasets, and a class token is added to the input image to produce the final classification result [34]. Though the transformer decoder is not used in the classification task, it can be adapted for dense prediction tasks like object detection and depth estimation. For example, detection transformer (DETR) [61] uses the transformer decoder for set prediction of objects in images, and MonoDTR [62] uses a depth-aware transformer decoder to achieve monocular 3-D object detection. A transformer decoder is also used in RS image segmentation tasks. For example, UnetFormer [63] uses a ResNet18 as the encoder and a transformer decoder to achieve real-time RS image segmentation. However, UnetFormer combines the Swin transformer and CNN for feature processing as previous Swin-transformer-based methods, which probably cannot fully exploit the global semantic information of RS images.

As mentioned above, most of the current transformer-based methods for RS image segmentation employ a transformer as either encoder or decoder, ignoring the potentiality of fully transformer-based methods. Furthermore, the common global-local feature extraction strategy that combines the Swin transformer and CNN is not suitable for RS image segmentation due to the large size of RS images. Therefore, in this article, a novel global-local transformer decoder is proposed to obtain coarse global and fine detailed information, which is more suitable for RS image segmentation.

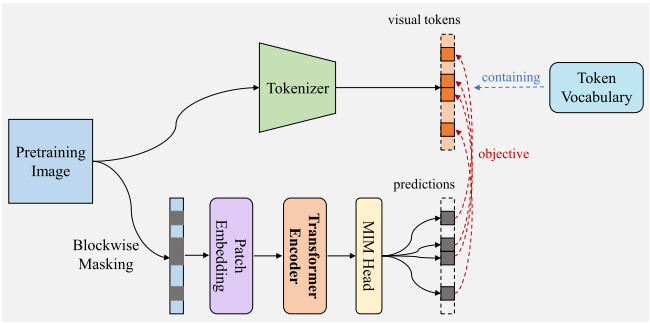
III. Methods

As shown in Fig. 2, the proposed GLOTS framework is established on the basis of an encoder–decoder structure which consists of a MIM transformer encoder and a global-local transformer decoder. The transformer decoder contains two main components: an FSAM and a GLAM. The MIM transformer encoder is capable of extracting semantic-rich features from RS images, which are then decoded by the global-local transformer decoder. In the decoding process, the FSAM separates the encoded features into multiscale ones and produces extra aggregation features. Then, the GLAM adequately utilizes the information of various scales and accomplishes global-local information acquisition and fusion. The details in upsampling are acquired by embedding the LPUS into the GLAM. An auxiliary loss is also introduced to accelerate the convergence of the model.

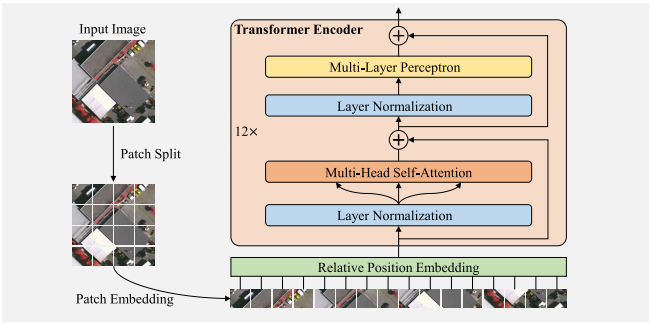
A. MIM Transformer Encoder

The proposed GLOTS employed the original ViT as the transformer encoder, which is pretrained with the MIM method. MIM [47] is a visual pretraining method that utilizes a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.



(a)



(b)

Fig. 3. Architecture of MIM-pretrained transformer encoder. (a) MIM pretraining process of the transformer encoder. (b) Internal structure of the transformer encoder.

self-masked pretraining mechanism inspired by the pretraining language model BERT [64]. MIM randomly masks parts of the input image to be pretrained and requires the network to reconstruct the original image, thereby achieving the pretraining of image data. Compared with the most commonly used Swin transformer, the MIM-pretrained transformer provides more semantic-rich representations while introducing less inductive bias, making it more suitable and flexible for large-scale RS image segmentation. As shown in Fig. 3(a), the MIM-pretraining process is composed of a Tokenizer, a transformer encoder, and a MIM Head. The input image $\boldsymbol{x}$ is first split into N image patches $\{\boldsymbol{x}_i^p\}_{i=1}^N$, and then the Tokenizer converts these image patches into a sequence of tokens $\{z_i\}_{i=1}^N$, which is achieved by the latent codes of discrete variational autoencoder (dVAE) [65]. The transformer encoder and the MIM Head are adopted to encode the embedded patches and make the score predictions, respectively. By acquiring the generated latent visual tokens and predictions, the MIM pretraining process, which aims at maximizing the log-likelihood of the correct visual tokens, can be formulated as follows:

$$\max \sum_{\boldsymbol{x}\in\mathcal{D}} \mathbb{E}_{\mathcal{M}} \left[\sum_{i\in\mathcal{M}} \log p_{\mathrm{MIM}}(z_i|\boldsymbol{x}^{\mathcal{M}}) \right]$$
$$\log p_{\mathrm{MIM}}(z_i|\boldsymbol{x}^{\mathcal{M}}) = \mathrm{softmax}_{z_i}(\boldsymbol{W}\boldsymbol{o}+\boldsymbol{b}) \quad (1)$$

where $\mathcal{D}$ is the training dataset, $\mathcal{M}$ is the set of positions of the masked patches, $\boldsymbol{x}^{\mathcal{M}} = \{\{\boldsymbol{x}_i^p\}_{i=1}^N : i \in \mathcal{M}\}$ denotes masked image patches, z_i is the latent visual token, $\boldsymbol{W}$ and $\boldsymbol{b}$ are parameters of the MIM Head, and $\boldsymbol{o}$ is the output of the transformer encoder [47].

The MIM-pretrained transformer encoder is used to extract semantic and appearance information of the input image, as shown in Fig. 3(b). Specifically, for a given RS image $\boldsymbol{x} \in \mathbb{R}^{H\times W\times C}$, it is divided into N patches with $P \times P$ pixels, that is $\boldsymbol{x}^p \in \mathbb{R}^{N\times(P^2\times C)}$. The patches are then projected into patch embedding $\boldsymbol{e}^p \in \mathbb{R}^{N\times D}$ by a patch embedding layer, where D is the embedding dimension. The patch embedding is combined with a relative position embedding to preserve the spatial positional information and then fed into the MIM-pretrained transformer encoder. Forward propagation of the transformer encoder is formulated as follows:

$$\boldsymbol{h} = \mathrm{MHSA}(\mathrm{LN}(\boldsymbol{e}^p + \boldsymbol{e}^{\mathrm{pos}})) + \gamma_1(\boldsymbol{e}^p + \boldsymbol{e}^{\mathrm{pos}})$$
$$\boldsymbol{o} = \mathrm{LN}(\mathrm{MLP}(\boldsymbol{h})) + \gamma_2(\boldsymbol{h}) \quad (2)$$

where $\boldsymbol{h}$ is a hidden feature of the transformer encoder, MHSA, MLP, and LN denote the multihead self-attention, multilayer perceptron, and layer normalization layers, respectively, and γ_1 and γ_2 are learnable parameters of the residual connections.

B. Global-Local Transformer Decoder

Visual transformers have been widely used in many computer vision tasks. However, most of the works only employ the pretrained transformers for feature extraction, totally ignoring the representation capability of transformers as downstream tasks. Though some works, e.g., UnetFormer [63], have tried to use a transformer as a decoder, they only combine the Swin transformer with CNNs, which is not suitable for large-scale RS image segmentation. In this work, a global-local transformer decoder is proposed to decode the hidden global and local information encoded by the MIM-pretrained transformer decoder. As shown in Fig. 2, the global-local transformer decoder is composed of an FSAM and a GLAM. By using transformers for both encoding and decoding in the network, consistent features can be acquired, which is beneficial for RS image segmentation.

1) Feature Separation-Aggregation Module: Different from the well-designed and highly-optimized CNNs, visual transformers generally lack multiscale feature processing modules like feature pyramid network (FPN) in CNNs. It is observed that single-scale feature processing can lead to severe information loss, especially for confused objects or inconspicuous objects. However, this problem can be alleviated by multiscale feature processing with a specific design (e.g., pyramid pooling module (PPM) [48]). Inspired by this, an FSAM is designed for multiscale feature acquisition, as shown in Fig. 4. The features output by the transformer encoder $\boldsymbol{o}$ are resized to four different scales, denoted as $\boldsymbol{f}_1$, $\boldsymbol{f}_2$, $\boldsymbol{f}_3$, and $\boldsymbol{f}_4$. Specifically, $\boldsymbol{f}_1$ and $\boldsymbol{f}_2$ are performed with a max pooling operation of $4\times$ and $2\times$ downsampling, respectively, $\boldsymbol{f}_3$ is the original feature, and $\boldsymbol{f}_4$ is obtained by a $2\times$ upsampling operation to integrate more fine details. The feature pyramid provides a multiscale feature representation for the subsequent global-local attention calculation. And this specific design is beneficial for the segmentation of complex objects in various scales, especially for small-scale objects. In addition, an extra aggregation branch is also adopted in the FSAM to fuse the multiscale features and parse semantic information of the encoded features (red line in Fig. 4).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.

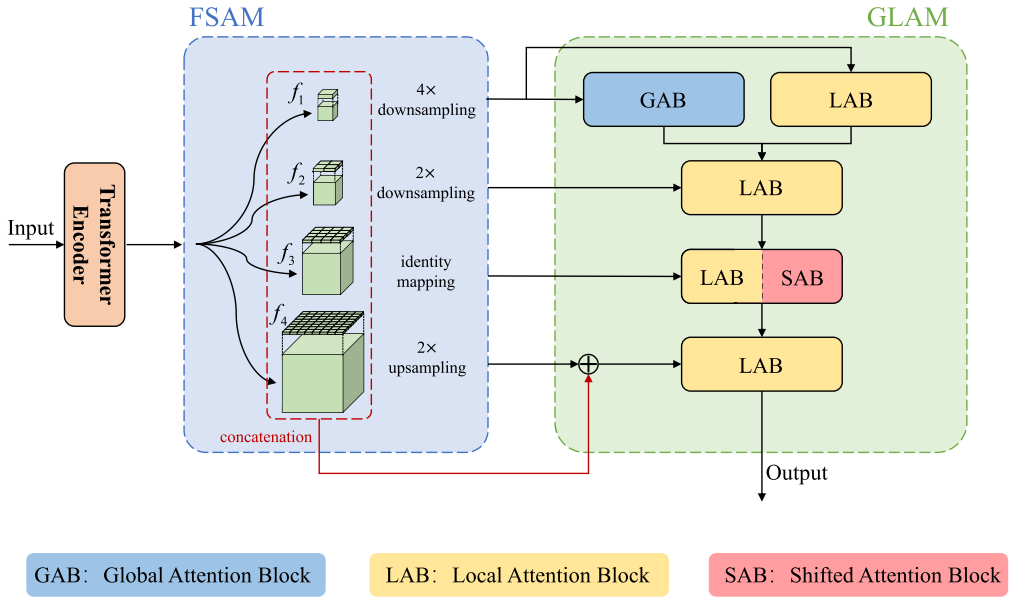


Fig. 4. Architecture of the global-local transformer decoder. It consists of an FSAM and a GLAM.

2) Global-Local Attention Module: CNN-based decoders have been proven to be effective for RS image segmentation, and UperNet [50] is a typical example. However, as transformer encoders are more flexible and scalable than CNN-based backbones, it is natural to use transformers as decoders. In this work, a GLAM is designed to acquire the global and local features for RS image segmentation. As shown in Fig. 4, the hierarchical features generated by FSAM are utilized by the corresponding attention blocks for global and local feature acquisition. Specifically, all the features are fed into a LAB for local attention computing, and the local attention maps are parsed progressively from the coarsest to the finest features. In addition, the coarsest global-level features are obtained by an extra GAB for global attention computing, and the attention maps are fused by a SAB to obtain the final feature representation.

Denoting the input features as $\boldsymbol{f_i} \in \mathbb{R}^{D\times(H_i\times W_i)}$, where $i \in \{1, 2, 3, 4\}$, the calculation within different blocks are shown in Fig. 5. The computational complexity of these three blocks is as follows:

$$\begin{aligned}\Omega(\text{GAB}) &= 4H_iW_iD^2 + 2(H_iW_i)^2D\\ \Omega(\text{LAB}) &= 4H_iW_iD^2 + 2M^2H_iW_iD\\ \Omega(\text{SAB}) &= \Omega(\text{LAB})\end{aligned} \tag{3}$$

where $(H_i \times W_i)$ denotes the number of patches in feature map $\boldsymbol{f_i}$ and M^2 denotes the number of patches in a window. It can be seen that the computational complexity of GAB is far more than that of LAB, especially when the input feature map is large. Therefore, though full global attention is demonstrated to be effective with sufficient training data and computational resources, LAB is still used to obtain the primary features because of the above-mentioned limitations. Inspired by [39], SAB is designed to facilitate formation interaction across the windows. The SAB cyclically shifts the windows toward the top-left direction, so that the attention used to be calculated across the original windows can be now calculated within the shifted windows, as shown in Fig. 5(c).

Specifically, as shown in Fig. 5, supposing feature map $\boldsymbol{F} \in \mathbb{R}^{H\times W}$ (considering the spatial dimension only) is divided into $K \times K$ windows, patch $\boldsymbol{p}_i$ and $\boldsymbol{p}_j$ are two patches in the same window, and $\boldsymbol{p}_k$ is in an adjacent window, $i, j, k \in \{1, 2, \ldots, K^2\}$. The attention map for $\boldsymbol{p}_i$ and $\boldsymbol{p}_j$ can be depicted by calculating the similarity between $\boldsymbol{p}_i$ and $\boldsymbol{p}_j$, which is

$$\boldsymbol{a}_{ij} = \text{softmax}\left(\boldsymbol{p}_i^T\boldsymbol{p}_j\right). \tag{4}$$

In this case, $\boldsymbol{p}_i$ and $\boldsymbol{p}_j$ have no interaction with $\boldsymbol{p}_k$. After shifting the windows, $\boldsymbol{p}_i$ and $\boldsymbol{p}_j$ are in different windows, and $\boldsymbol{p}_k$ is in the same window with $\boldsymbol{p}_j$. The attention map for $\boldsymbol{p}_j$ and $\boldsymbol{p}_k$ can be calculated as

$$\boldsymbol{a}_{jk} = \text{softmax}\left(\boldsymbol{p}_j^T\boldsymbol{p}_k\right). \tag{5}$$

Note that $\boldsymbol{p}_j$ in the shifted window used for calculating $\boldsymbol{a}_{jk}$ contains the information of $\boldsymbol{p}_i$ in the original window, which means that $\boldsymbol{p}_i$ and $\boldsymbol{p}_k$ also have interaction through $\boldsymbol{p}_j$.

However, these indirect interactions are not so effective, therefore the GAB is designed to calculate the direct global dependencies on the coarsest feature map, which is similar to the global average pooling (GAP) in scene understanding. The GAB is capable of computing each patch's global dependencies in the entire feature map $\boldsymbol{F} \in \mathbb{R}^{H\times W}$, implanting the global information into the local features, serving as a supplement to the local-level features.

C. Learnable Progressive Upsampling Strategy

The proposed GLOTS employs a LPUS, containing learnable upsampling and progressive upsampling. Upsampling is an essential step in semantic segmentation to restore the resolution reduced in encoding downsampling. To acquire more contextual information, models usually involve representations

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.

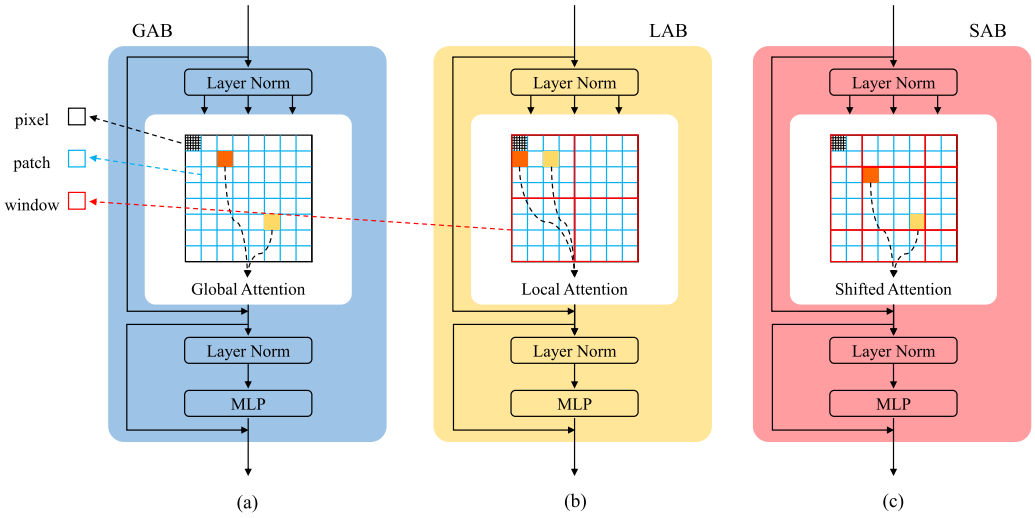


Fig. 5. Architecture of three attention blocks. (a) GAB. (b) LAB. (c) SAB.

from different scales in upsampling, and multiscale strategies have been widely considered to address this issue. As shown in Fig. 6(a), the multiscale fusion upsampling (MSFU) strategy concatenates the representations from different scales and then performs upsampling [18], [48]. However, the upsampled features are separated from the subsequent decoding process, which may lead to the loss of fine-grained details.

Inspired by [58], a multiscale progressive upsampling (MSPU) strategy integrating progressive upsampling strategy is constructed to aggregate contextual information from different scales while maximally alleviating the adversarial effect of upsampling, as shown in Fig. 6(b). The MSPU strategy uses multiscale features generated by FSAM to perform upsampling, and the concatenated features of the same scale are processed in the subsequent transformer decoder layer for decoding.

On the basis of MSPU, the LPUS is proposed to adopt a learnable transposed convolution operation, replacing the bilinear interpolation in the upsampling process. The transposed convolution operation can be regarded as a fractionally-strided convolution. Namely, a $2\times$ upsampling operation can be achieved by convolution with a stride of $1/2$. This LPUS is more flexible and adaptive than the previous upsampling strategies, and it can be trained together with the Global-Local transformer decoder to recover the coarse semantic features and fine-grained details of the original images.

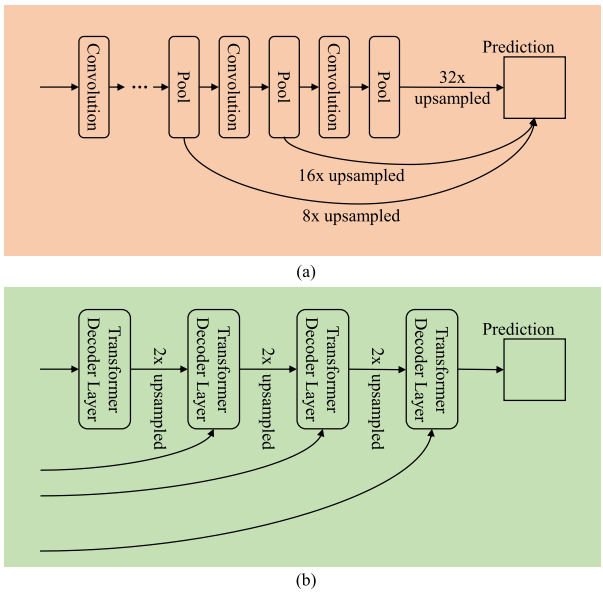


Fig. 6. Two different upsampling strategies. (a) Multiscale fusion upsampling. (b) Multiscale progressive upsampling.

D. Loss Function

An auxiliary loss is commonly adopted to accelerate the convergence of the model and improve the performance. In this work, the cross-entropy loss is used as the primary loss function and focal loss [66] is used as the auxiliary loss. The focal loss is a variant of the cross-entropy loss designed to alleviate the imbalance problem of the training set, and it is beneficial for hard sample mining. The objective function of the model is thus defined as follows:

$$\mathcal{L} = \mathcal{L}_{\text{main}} + \lambda \mathcal{L}_{\text{aux}} \tag{6}$$

where $\mathcal{L}_{\text{main}}$ is the cross-entropy loss, $\mathcal{L}_{\text{aux}}$ is the focal loss, and λ is the balance coefficient of auxiliary loss.

IV. Experiments

A. Datasets

The experiments are conducted on three public high-resolution RS datasets, including the ISPRS Vaihingen dataset, the ISPRS Potsdam dataset [40], and the LoveDA dataset [41].

1) Vaihingen Dataset: The Vaihingen dataset is a widely used benchmark for semantic segmentation of high-resolution RS images provided by the International Society for Photogrammetry and RS (ISPRS) and released in 2014. The dataset contains high-resolution aerial images of the town of Vaihingen, Germany, and manually annotated ground truth data for six different land cover classes: impervious surface, building, low vegetation, tree, car, and clutter/background.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.

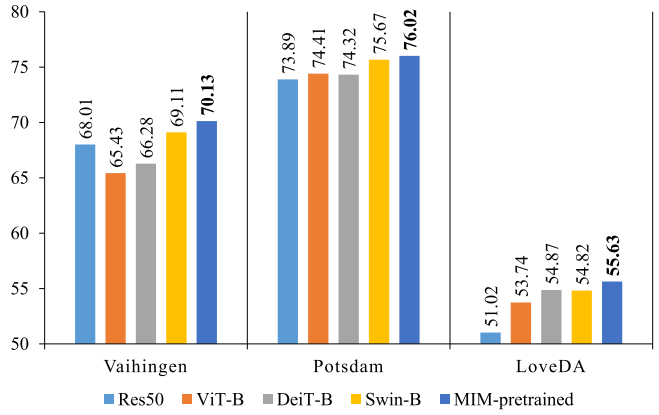


Fig. 7. Results of different encoders on the three RS datasets (in mIoU%).

These 33 true orthophotos (TOPs) images own a spatial resolution of 0.09 m/pixel and with sizes ranging from 1887 to 3816 pixels in width. In order to evaluate the performance of the proposed method, the standard protocol is followed to divide the dataset into a training set with 16 images and a testing set with 17 images. For convenience, the TOP images are cropped into 512 × 512 pixels patches with a stride of 256 pixels.

2) Potsdam Dataset: The Potsdam dataset is another widely used benchmark for semantic segmentation of high-resolution RS images, released in 2018. The dataset contains high-resolution aerial images of the city of Potsdam, Germany, and it has the same manually annotated classes as the Vaihingen dataset. The images are acquired by a Leica ADS80 airborne sensor and cover an area of 38.4 km^2 at a spatial resolution of 0.05 m. Compared with the Vaihingen dataset, the Potsdam dataset has a larger scale with 38 TOP images and a larger image size with 6000 × 6000 pixels. Following the common practice, the dataset is divided into a training set of 24 images and a test set of 14 images, and the same cropping strategy is adopted as the Vaihingen dataset.

3) LoveDA Dataset: The LoveDA dataset is a newly released dataset for semantic segmentation of high-resolution RS images (released in 2021) which is more challenging because of its complex background samples, various object scales, and inconsistent class distributions. The dataset contains 5987 images that have already been cropped into patches (each patch contains 1024 × 1024 pixels), with seven annotated classes including building, road, water, vegetation, car, clutter, and background. The covered area of the LoveDA dataset is 536.15 km^2, and the ground sampling distance (GSD) is 0.3 m. Following the official dataset split, 2522 images are used for training and 1669 images for testing in the experiments.

B. Evaluation Metrics

In this work, two commonly used metrics are used to evaluate the performance evaluation of the proposed method, including the mean intersection over union (mIoU) and the mean accuracy (mAcc).

1) Mean Intersection Over Union: The mIoU is one of the primary metrics for the performance of semantic segmentation models, which is defined as the average of the IoU of each class. Assuming that true positive (TP), false positive (FP), and false negative (FN) represent the number of pixels that are correctly, incorrectly, and missed classified, respectively, the mIoU is then defined as

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N}\frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i} \tag{7}$$

where N is the number of classes.

2) Mean Accuracy: The mAcc is another effective metric for segmentation performance evaluation, defined as the average of the accuracy of each class

$$\text{mAcc} = \frac{1}{N}\sum_{i}^{N}\frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i}. \tag{8}$$

C. Experiment Settings

To balance the performance and efficiency, the input image size is set to 512 × 512 pixels, and the batch size is set to 8. The Adaptive momentum estimation with Weight decay (AdamW) optimizer is used for model optimization, in which the learning rate is set to 3×10^{-5}, and weight decay is set to 0.05. A polynomial learning rate scheduler is used for the learning rate decaying after a warm-up period of 1500 iterations, and the total training iterations are set to 160 000. Multiscale training and testing (MSTrTe) data augmentation, such as random flipping, random rotation, random scaling, and random cropping, is used to improve the generalization ability of the model. The training process is conducted on four NVIDIA RTX 3090 GPUs, each with 24 GB memory under the PyTorch framework. All the experiments are conducted on a Linux Ubuntu 20.04 system.

D. Ablation Study

1) Comparisons of Different Encoders: ViT, with sufficient data and computing resources, can achieve performance improvements compared with CNNs. However, the performance of ViT is not robust when applied to small datasets, even with the help of pretrained models. Efforts have been made to improve the robustness of ViT, such as distilling the knowledge from CNNs [35] and introducing model prior knowledge of CNNs [39]. In this work, the MIM-pretrained transformer encoder is adopted as it can obtain more semantic information from the input image. In order to verify the effectiveness of the MIM-pretrained transformer encoder, it is compared with some other commonly used encoders, including ResNet50, ViT, DeiT, and Swin. The proposed methods employing different encoders are shown in Fig. 7. It can be seen that the basic ViT encoder performs inferiorly to ResNet50 on the Vaihingen dataset, while in other cases, visual transformers generally outperform CNNs on the three RS datasets. Particularly, the MIM-pretrained ViT encoder achieves the best performance on the three datasets, which gains a slight improvement over the widely adopted

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.

TABLE I
RESULTS OF DIFFERENT UPSAMPLING STRATEGIES ON THE ISPRS VAIHINGEN DATASET

Strategies	Operation			IoU(%)						Evaluation metrics	
	Multi-scale fusion	Multi-scale progressive	Learnable upsampling	Impervious surface	Building	Low vegetation	Tree	Car	Clutter	mIoU(%)	mAcc(%)
DirectU				81.04	88.74	67.8	76.02	58.39	29.73	66.95	75.52
MFSU	✓			82.28	89.14	68.78	76.47	65.63	30.89	68.87	76.95
LMFU	✓		✓	82.28	88.97	68.57	76.53	**67.74**	30.26	69.06	76.94
MSPU		✓		81.59	87.54	67.89	76.39	66.14	34.88	69.07	77.17
LPUS		✓	✓	**82.65**	**89.17**	**69.36**	**77.04**	67.55	**35**	**70.13**	**77.81**

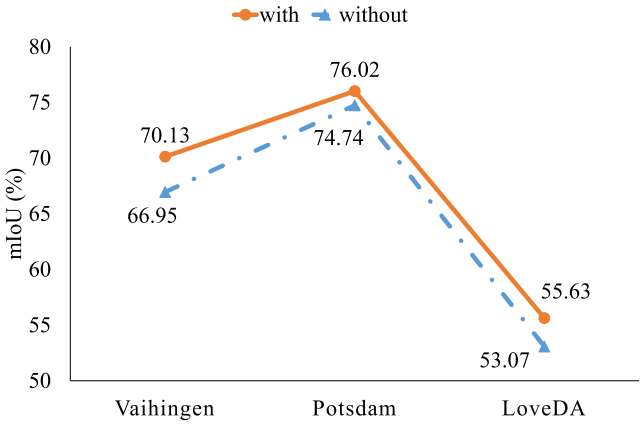


Fig. 8. Results of the proposed method with or without FSAM on the three RS datasets.

Swin transformer encoder for RS image segmentation, demonstrating the effectiveness of the MIM-pretrained transformer encoder in the proposed GLOTS.

2) Effectiveness of the Global-Local Transformer Decoder: To verify the effectiveness of the proposed global-local transformer decoder, two individual experiments are conducted to explore the effectiveness of the FSAM and the GLAM, respectively.

As shown in Fig. 8, without the FSAM, the performance of the proposed method is degraded over 1.28% mIoU on the three datasets, which demonstrates the significance of the multiscale strategy for RS image segmentation. This can also be verified by the visualization results in Fig. 9, where the proposed method with FSAM obtains more accurate segmentation results than that without FSAM, especially for the small objects (e.g., cars).

There are three variants of the GLAM, including using the GAB, using the LAB, and using the aggregation of GAB and LAB (GAB + LAB) on the most coarse scale due to the memory limitation. As shown in Fig. 10, when employing both GAB and LAB, the proposed method can achieve the best performance. Without GAB, the performance is degraded which indicates that global information acquisition is beneficial for the performance of the proposed method. It is also notable that without LAB, the performance is also degraded obviously, which may be because local dependencies are also important in feature map decoding.

3) Comparisons of Different Upsampling Methods: In order to verify the effectiveness of the proposed LPUS, experiments are conducted to compare the performance of the proposed LPUS with direct upsampling (DirectU), MFSU, learnable multiscale fusion upsampling (LMFU), and MSPU, as shown in Table I. The DirectU denotes directly upscaling the outputs of the global-local transformer decoder to prediction maps, and it is achieved by removing the FSAM. These five upsampling strategies are actually different combinations of three different operations (multiscale fusion, MSPU, and learnable upsampling). Improved segmentation performance can be observed by employing multiscale strategies which contain multiscale fusion or MSPU, while MSPU is slightly more effective than fusion. The performance can be further improved by adopting the learnable upsampling strategy. As a result, the proposed LPUS, which is actually the combination of MSPU and learnable upsampling, achieves the best results. It is notable that the mIoU of the class car is significantly degraded when the multiscale strategies are not employed, which also demonstrates that it is necessary to fuse the multiscale features for RS image segmentation, especially for small objects.

E. Comparison With State-of-the-Art RS Image Segmentation Methods

In order to further evaluate the effectiveness of the proposed GLOTS, it is compared with several state-of-the-art RS image segmentation methods, including CNN-based methods such as FCN [18], U-Net [19], PSPNet [48], DeepLabV3+ [51], BiSeNet [67], HRNet [21], as well as transformer-based methods such as SETR [58], SegFormer [60], BeiT [47], Swin transformer [39], Swin-Unet [68], DC-Swin [44], ST-UNet [42], SwinB-CNN [43], and UnetFormer [63]. These methods are implemented using the same data preprocessing and training settings as GLOTS. It is worth noting that the previous works tend to abandon the clutter class in their experiments on the ISPRS datasets. However, the clutter class is also vital for the RS segmentation task to evaluate the background perception capabilities of different methods. In this work, all the methods are retrained and evaluated on the three RS datasets to include the clutter class. Some of their results could be different from those reported in the original papers due to the inconsistency of the training settings and data split. Pretrained models of the methods are acquired from the official websites or the PyTorch Image Models (Timm) library.

The comparison results on the Vaihingen dataset and the Potsdam dataset exhibit many similarities, which are shown in Tables II and III, respectively. The computational complexity of different methods is also listed in Table II using the inference time of a single image with the size of 512 × 512 as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON WITH SOME STATE-OF-THE-ART METHODS ON THE ISPRS VAIHINGEN DATASET

Applications	Structures	Methods	Backbones	IoU(%)						Evaluation metrics		Inference time (FPS)
				Impervious surface	Building	Low vegetation	Tree	Car	Clutter	mIoU(%)	mAcc(%)	
natural scene	CNN-based	FCN [18]	ResNet50	80.42	87.8	64.85	74.45	60.03	35.15	67.11	75.2	46.9
		U-Net [19]	U-Net	77.68	83.69	62.43	73.46	55.49	12.14	60.82	69.87	16.1
		PSPNet [48]	ResNet50	80.73	88.11	65.74	75.02	65.39	31.17	67.69	75.92	47.6
		DeepLabV3+ [51]	ResNet50	79.74	87.24	64.86	74.65	61.05	**40.21**	67.96	77.1	42.1
		BiSeNet [68]	ResNet18	78.77	85.24	63.97	73.54	52.39	31.54	64.24	73.47	**127.1**
		HRNet [21]	HRNetv2p-W18	81.15	88.16	65.55	74.78	65.44	33.41	68.08	76.4	45.2
	Trans En	SETR [58]	ViT-B	80.27	87.73	67.21	74.95	52.4	23.57	64.36	72.85	13.5
		SegFormer [60]	MiT-B5	81.9	88.69	67.5	76.24	66.13	28.61	68.18	76.21	23.8
		BeiT [47]	ViT-B	82.16	88.61	68.87	76.77	65.86	35.26	69.59	77.45	10
		Swin Transformer [39]	Swin-B	81.8	88.6	68.38	76.58	61.91	35.3	68.76	76.71	25.3
	Trans En & De	Swin-Unet [69]	SwinUnet-B	77.86	80.78	65.71	68.75	37.42	19.23	58.29	68.2	17.2
RS scene	Trans En	ST-Unet [42]	Swin-B+CNN	82.07	89.22	67.81	76.27	65.03	35.68	69.35	77.24	21.2
		SwinB-CNN [43]	Swin-B	81.66	**89.32**	67.59	75.86	63.88	34.41	68.79	76.84	23.3
		DC-Swin [44]	Swin-B	80.98	88.39	66.8	75.65	63.18	31.72	67.79	76.2	24.1
	Trans De	UnetFormer [64]	ResNet18	79.87	86.68	65.1	74.21	59.42	27.55	65.47	74.06	115.3
	Trans En & De	GLOTS	ViT-B	**82.65**	89.17	**69.36**	**77.04**	**67.55**	35	**70.13**	**77.81**	19.8

TABLE III
COMPARISON WITH SOME STATE-OF-THE-ART METHODS ON THE ISPRS POTSDAM DATASET

Applications	Structures	Methods	Backbones	IoU(%)						Evaluation metrics	
				Impervious surface	Building	Low vegetation	Tree	Car	Clutter	mIoU(%)	mAcc(%)
natural scene	CNN-based	FCN [18]	ResNet50	82.16	89.94	71.26	74.38	82.41	33.74	72.31	81
		U-Net [19]	U-Net	80.36	87.03	69.76	72.34	81	25.15	69.28	78.15
		PSPNet [48]	ResNet50	83.07	91.15	71.92	73.87	81.31	39.21	73.42	82.05
		DeepLabV3+ [51]	ResNet50	83.59	91.08	72.6	75.4	82.95	38.14	73.96	82.48
		BiSeNet [68]	ResNet18	80.23	86.88	70.37	71.76	78.31	35.92	70.58	80.33
		HRNet [21]	HRNetv2p-W18	83.34	90.85	72.36	74.59	82.13	35.52	73.13	81.63
	Trans En	SETR [58]	ViT-B	83.35	91.52	73.53	75.21	79.55	38.73	73.65	82.44
		SegFormer [60]	MiT-B5	84.81	91.94	74.19	75.93	83.61	39.08	74.93	82.67
		BeiT [47]	ViT-B	84.41	92.11	74.76	76.34	84.01	38.62	75.04	82.91
		Swin Transformer [39]	Swin-B	84.24	92.04	74.98	76.32	82.34	40.54	75.08	82.96
	Trans En & De	Swin-Unet [69]	SwinUnet-B	73.74	74.41	65	56.39	57.6	15.95	57.18	73.7
RS scene	Trans En	ST-Unet [42]	Swin-B+CNN	84.72	92.13	74.73	74.95	81.83	40.43	74.8	83.19
		SwinB-CNN [43]	Swin-B	84.52	**92.4**	75.07	76.88	82.66	39.06	75.1	82.79
		DC-Swin [44]	Swin-B	84	91.84	73.87	75.49	81.68	40.03	74.48	82.79
	Trans De	UnetFormer [64]	ResNet18	82.15	90.11	71.85	73.5	80.13	37.39	72.52	81.36
	Trans En & De	GLOTS	ViT-B	**85.2**	92.19	**75.23**	**76.97**	**84.37**	**42.15**	**76.02**	**83.66**

the metric (FPS), measured on a single NVIDIA GeForce RTX 3090 GPU. The inference time is more intuitive, practical, and widely used, compared with other metrics such as FLOPs and parameters. It can be observed that the segmentation accuracy on the Potsdam dataset is generally higher than that on the Vaihingen dataset, which can be probably attributed to the larger dataset size of the Potsdam dataset. Particularly, the transformer-based methods show greater improvements than the CNN-based methods on the Potsdam dataset, which indicates the flexibility and scalability of the transformer. However, on both two datasets, CNN-based methods such as FCN, PSPNet, DeepLabV3+, and HRNet achieve competitive performance with the highly optimized ResNet50 backbone embedded with dilated convolutions [69]. The transformer-based methods which employ enhanced visual transformers, such as SegFormer, BeiT, and Swin transformer, also achieve promising performance. In contrast, SETR, which adopts the vanilla ViT, is slightly inferior to these methods, especially

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON WITH SOME STATE-OF-THE-ART METHODS ON THE LOVEDA DATASET

Applications	Structures	Methods	Backbones	IoU(%)							Evaluation metrics	
				background	building	road	water	barren	forest	agricultural	mIoU(%)	mAcc(%)
natural scene	CNN-based	FCN [18]	ResNet50	50.63	63.38	53.09	60.57	24.36	38.45	47.33	48.26	61.8
		U-Net [19]	U-Net	50.1	56.44	47.38	47.96	22.57	40.16	42.18	43.83	55.94
		PSPNet [48]	ResNet50	52.73	62.75	51.33	57.48	23.06	36.43	48.21	47.43	58.32
		DeepLabV3+ [51]	ResNet50	53.98	63.87	54.05	62.07	29.24	38.59	49.45	50.18	62.79
		BiSeNet [68]	ResNet18	50.56	53.91	49.83	57.96	18.7	40.17	45.47	45.23	57.69
		HRNet [21]	HRNetv2p-W18	51.61	61.99	52.56	67.05	27.65	39.47	50.87	50.17	62.19
	Trans En	SETR [58]	ViT-B	53	66.3	57.13	71.1	29.26	42.97	57.54	53.9	65.83
		SegFormer [60]	MiT-B5	54.69	65.94	54.03	70.74	31.07	37.82	53.66	52.56	63.7
		BeiT [47]	ViT-B	53.01	63.95	57.79	71.43	29.36	**45.15**	58.33	54.15	66.58
		Swin Transformer [39]	Swin-B	54.6	65.12	57.84	72.86	32.27	44.72	52.77	54.31	67.13
	Trans En & De	Swin-Unet [69]	SwinUnet-B	44.64	51.73	34.92	53.97	15.43	28.73	38.15	38.37	48.57
RS scene	Trans En	ST-Unet [42]	Swin-B+CNN	53.75	65.84	57.85	72.35	30.23	44	56.98	54.43	67.18
		SwinB-CNN [43]	Swin-B	53.6	65.98	58.36	72.67	31.6	43.56	53.62	54.2	66.9
		DC-Swin [44]	Swin-B	48.14	65.14	55.4	70.23	**39.38**	38.01	59.46	53.68	**67.68**
	Trans De	UnetFormer [64]	ResNet18	42.52	60.04	50.97	55.52	6.34	35.12	46.08	42.37	54.51
	Trans En & De	GLOTS	ViT-B	**55.27**	**67.06**	**59.61**	**73.08**	29.18	45.09	**60.09**	**55.63**	67.42

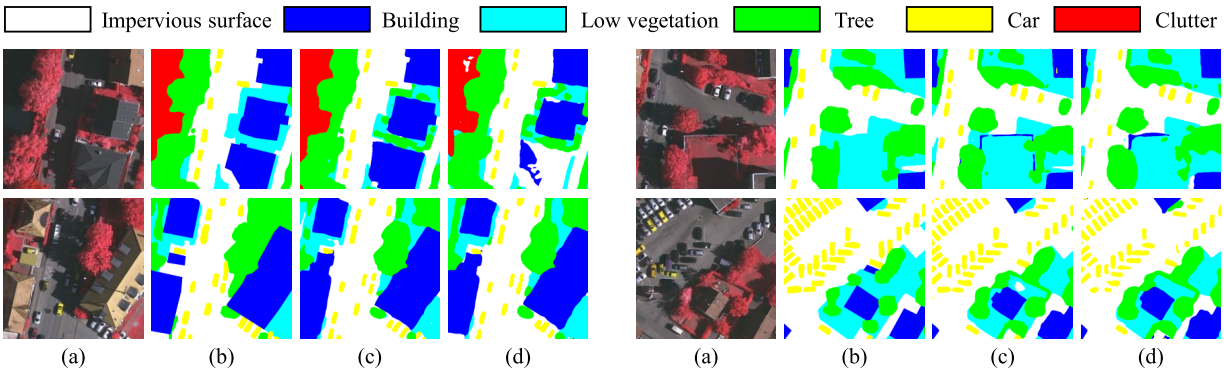


Fig. 9. Visualization results of the proposed method with or without FSAM on the three RS datasets. (a) Image. (b) Ground truth. (c) With FSAM. (d) Without FSAM.

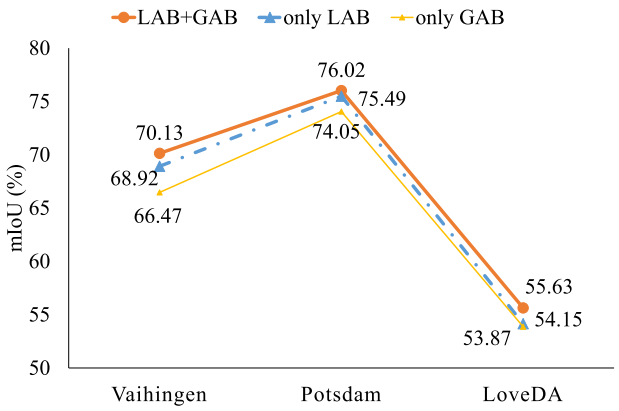


Fig. 10. Results of the proposed method with three variants of GLAM on the three RS datasets.

on the Vaihingen dataset, probably due to the limited dataset size. However, this issue is significantly alleviated on the Potsdam dataset, probably because sufficient training images are available. In addition, the Swin transformer-based methods, such as ST-Unet, SwinB-CNN, and DC-Swin, are capable of acquiring satisfactory results on the two ISPRS datasets. It is notable that two real-time methods, BiSeNet and UnetFormer, can also achieve competitive performance though they are slightly inferior to the other methods. However, the performances of U-Net and Swin-Unet are relatively poor, probably because U-Net cannot be optimized with dilated-ResNet50 and retains a simple backbone, while Swin-Unet is designed with a fixed input size of 224 × 224, which is not compatible with the RS images and the backbone is not the same as Swin transformer to load the pretrained weights. Overall, the proposed GLOTS can achieve the highest mIoU of 70.13% and mAcc of 77.81% on the Vaihingen dataset and 76.02% in mIoU and 82.54% in mAcc on the Potsdam dataset, respectively, outperforming the other methods on most of the classes except for slight inferiority in the building class on the two datasets and the clutter class on the Vaihingen dataset. This is probably because of the relatively simple structure of the building class that is easier to be recognized by other methods, and the insufficient training images of the clutter

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.

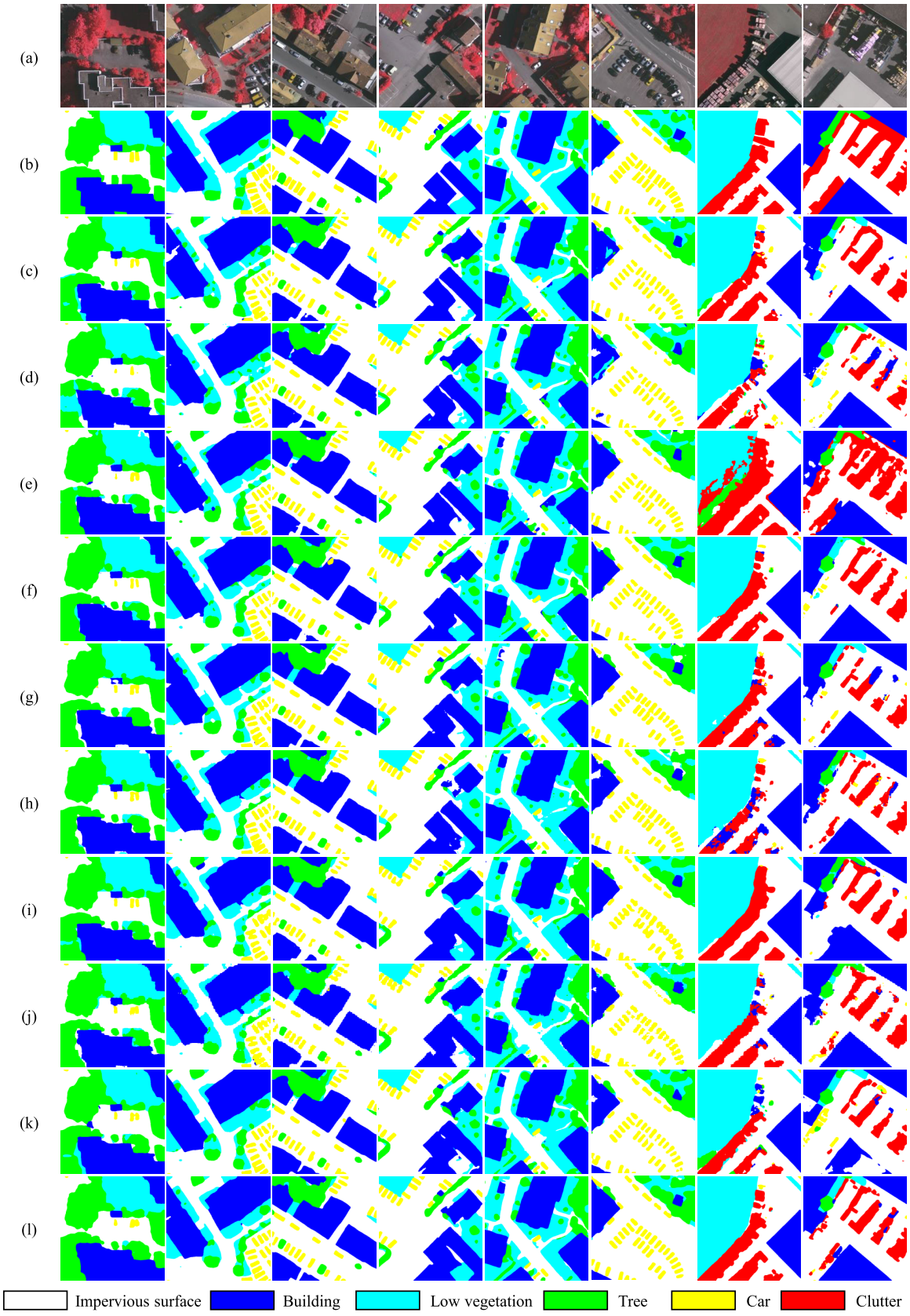


Fig. 11. Visualization results on the ISPRS Vaihingen dataset. (a) Image. (b) GT. (c) FCN. (d) U-Net. (e) Deeplabv3+. (f) HRNet. (g) SegFormer. (h) BeiT. (i) Swin Transformer. (j) DC-Swin. (k) UnetFormer. (l) GLOTS.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.

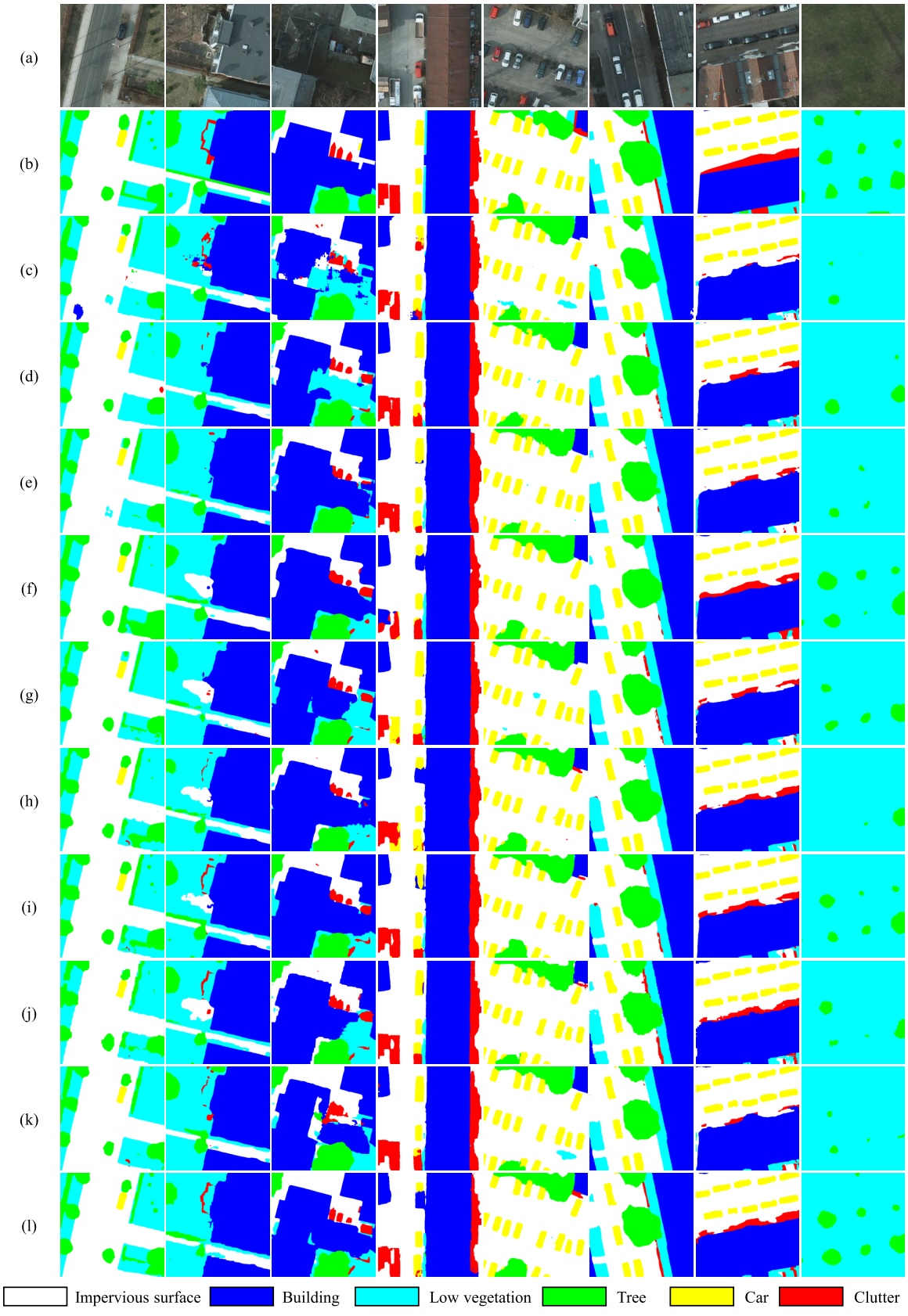


Fig. 12. Visualization results on the ISPRS Potsdam dataset. (a) Image. (b) GT. (c) U-Net. (d) Deeplabv3+. (e) HRNet. (f) SETR (g) SegFormer. (h) ST-Unet (i) SwinB-CNN. (j) DC-Swin. (k) UnetFormer. (l) GLOTS.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.

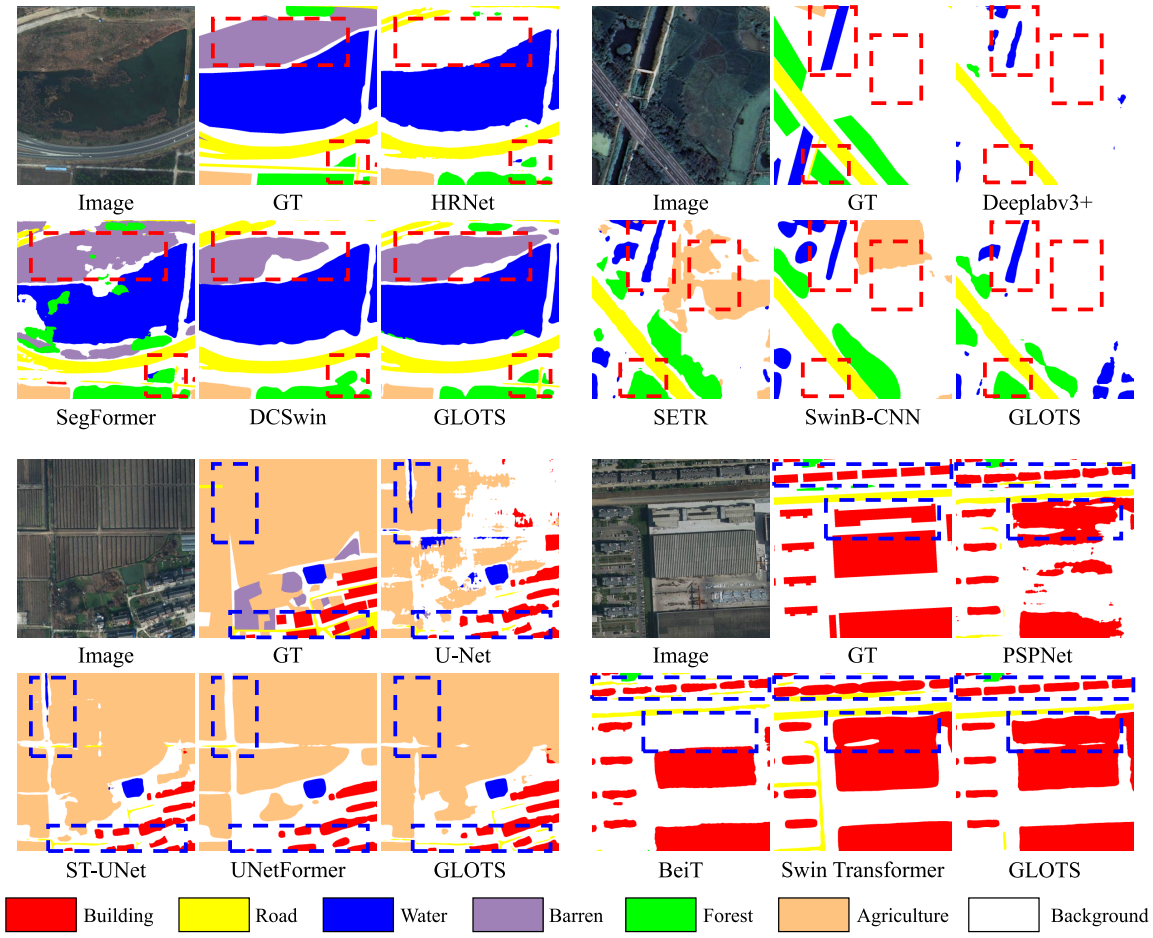


Fig. 13. Detailed comparisons of visualization results on the LoveDA dataset.

class in the Vaihingen dataset to which the transformer-based methods are generally sensitive. The inference time of the proposed GLOTS is comparable to that of the transformer encoder methods, while generally slower than the CNN-based methods. The main reason is that the transformer structure is more complex than the CNN structure. The two real-time methods, namely, BiSeNet and UnetFormer, are noticeably faster than the other methods, with more lightweight encoders. The visualization results on the two datasets are shown in Figs. 11 and 12, respectively, demonstrating that GLOTS can achieve more accurate segmentation results than the other methods.

The LoveDA dataset contains more classes than the previous two datasets, which makes it more challenging and difficult for accurate segmentation. The comparison results on the LoveDA dataset are shown in Table IV. It can be seen that the performance of the transformer-based methods surpasses the CNN-based methods by a larger margin compared with the Potsdam dataset. This is probably because the LoveDA dataset is more complex and owns more sufficient images than the Potsdam dataset. The proposed GLOTS significantly outperforms the other methods in mIoU and is only slightly inferior to the best method (DC-Swin) in mAcc. Detailed comparison of visualization results on this challenging dataset is shown in Fig. 13, in which red and blue boxes denote the superiority in detail of GLOTS over other methods. It is obvious that though the sampling objects and background are complex and diverse, the proposed GLOTS can still acquire both semantic-rich and detail-rich features, and hence achieve better segmentation results. Specifically, the proposed GLOTS can accurately segment the edges of the buildings and trees, and the boundaries of the roads and water bodies, which are difficult for other methods.

V. Conclusion

In this work, a novel GLOTS is proposed to address the issue that RS segmentation needs both global semantic information and local detail information. GLOTS employs a MIM-pretrained transformer encoder to acquire semantic-rich information and a global-local transformer decoder to decode coarse and fine features. The crucial components of GLOTS, including the FSAM and GLAM, are demonstrated to be effective in experiments with concrete ablation studies. Furthermore, the entire framework of GLOTS is proved to be superior to the state-of-the-art methods on the LoveDA and ISPRS datasets with quantitative and qualitative evaluations.

In the future, more attempts will be made in designing more efficient transformer-based RS segmentation methods, such as full global attention transformer with sparse connections, and improving the performance of transformer-based RS segmentation methods in label-insufficient scenarios, such as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.

semisupervised learning and weakly-supervised learning with GLOTS.

REFERENCES

[1] D. Maktav, F. Erbek, and C. Jürgens, "Remote sensing of urban areas," *Int. J. remote Sens.*, vol. 26, no. 4, pp. 655–659, 2005.

[2] M. Kampffmeyer, A.-B. Salberg, and R. Jenssen, "Semantic segmentation of small objects and modeling of uncertainty in urban remote sensing images using deep convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2016, pp. 680–688.

[3] S. Ali Khan, Y. Shi, M. Shahzad, and X. X. Zhu, "FGCN: Deep feature-based graph convolutional network for semantic segmentation of urban 3D point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 778–787.

[4] T. P. P. Sharma, J. Zhang, U. A. Koju, S. Zhang, Y. Bai, and M. K. Suwal, "Review of flood disaster studies in Nepal: A remote sensing perspective," *Int. J. Disaster Risk Reduction*, vol. 34, pp. 18–27, Mar. 2019.

[5] C. Van Westen, "Remote sensing for natural disaster management," *Int. Arch. Photogramm. remote Sens.*, vol. 33, pp. 1609–1617, Jul. 2000.

[6] M. Wieland, S. Martinis, and Y. Li, "Semantic segmentation of water bodies in multi-spectral satellite images for situational awareness in emergency response," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, pp. 273–277, Sep. 2019.

[7] J. Li, Y. Pei, S. Zhao, R. Xiao, X. Sang, and C. Zhang, "A review of remote sensing for environmental monitoring in China," *Remote Sens.*, vol. 12, no. 7, p. 1130, Apr. 2020.

[8] M. V. C. V. D. Costa et al., "Remote sensing for monitoring photovoltaic solar plants in Brazil using deep semantic segmentation," *Energies*, vol. 14, no. 10, p. 2960, May 2021.

[9] B. Zhang, D. Wu, L. Zhang, Q. Jiao, and Q. Li, "Application of hyperspectral remote sensing for environment monitoring in mining areas," *Environ. Earth Sci.*, vol. 65, no. 3, pp. 649–658, Feb. 2012.

[10] Y. Liu, Q. Ren, J. Geng, M. Ding, and J. Li, "Efficient patch-wise semantic segmentation for large-scale remote sensing images," *Sensors*, vol. 18, no. 10, p. 3232, Sep. 2018.

[11] B. Neupane, T. Horanont, and J. Aryal, "Deep learning-based semantic segmentation of urban features in satellite images: A review and meta-analysis," *Remote Sens.*, vol. 13, no. 4, p. 808, Feb. 2021.

[12] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 166–177, Jun. 2019.

[13] F. I. Diakogiannis, F. Waldner, P. Caccetta, and C. Wu, "ResUNet-A: A deep learning framework for semantic segmentation of remotely sensed data," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 94–114, Apr. 2020.

[14] I. Kotaridis and M. Lazaridou, "Remote sensing image segmentation advances: A meta-analysis," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 309–322, Mar. 2021.

[15] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607713.

[16] M. Neubert, H. Herold, and G. Meinel, "Assessing image segmentation quality—Concepts, methods and application," in *Object-Based Image Analysis* (Lecture Notes in Geoinformation and Cartography). Berlin, Germany: Springer, 2008, pp. 769–784.

[17] X. Zheng, X. Wu, L. Huan, W. He, and H. Zhang, "A gather-to-guide network for remote sensing semantic segmentation of RGB and auxiliary image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5404915.

[18] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[19] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.

[20] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*. Berlin, Germany: Springer, 2018, pp. 3–11.

[21] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.

[22] H. Li, K. Qiu, L. Chen, X. Mei, L. Hong, and C. Tao, "SCAttNet: Semantic segmentation network with spatial and channel attention mechanism for high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 905–909, May 2021.

[23] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[24] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.

[25] Z. Xu, W. Zhang, T. Zhang, and J. Li, "HRCNet: High-resolution context extraction network for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 13, no. 1, p. 71, Dec. 2020.

[26] W. Sun and R. Wang, "Fully convolutional networks for semantic segmentation of very high resolution remotely sensed images combined with DSM," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 3, pp. 474–478, Mar. 2018.

[27] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[28] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.

[29] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[30] Y. Ding et al., "AF2GNN: Graph convolution with adaptive filters and aggregator fusion for hyperspectral image classification," *Inf. Sci.*, vol. 602, pp. 201–219, Jul. 2022.

[31] Z. Zhang et al., "Multireceptive field: An adaptive path aggregation graph neural framework for hyperspectral image classification," *Exp. Syst. Appl.*, vol. 217, May 2023, Art. no. 119508.

[32] Y. Ding et al., "Multi-scale receptive fields: Graph attention neural network for hyperspectral image classification," *Exp. Syst. Appl.*, vol. 223, Aug. 2023, Art. no. 119858.

[33] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1–11.

[34] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent.*, 2021, pp. 1–22.

[35] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jegou, "Training data-efficient image transformers & distillation through attention," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, Jul. 2021, pp. 10347–10357.

[36] B. Graham et al., "LeViT: A vision transformer in ConvNet's clothing for faster inference," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 12239–12249.

[37] X. Chu et al., "Twins: Revisiting the design of spatial attention in vision transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 9355–9366.

[38] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 548–558.

[39] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[40] ISPRS. (2018). *2D Semantic Labeling Dataset*, Accessed: Jun. 1, 2023. [Online]. Available: https://www2.isprs.org/commissions/comm2/wg4/benchmark/semantic-labeling/

[41] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.

[42] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.

[43] C. Zhang, W. Jiang, Y. Zhang, W. Wang, Q. Zhao, and C. Wang, "Transformer and CNN hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408820.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.

[44] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[45] Z. Xu, W. Zhang, T. Zhang, Z. Yang, and J. Li, "Efficient transformer for remote sensing image segmentation," *Remote Sens.*, vol. 13, no. 18, p. 3585, Sep. 2021.
[46] L. Gao et al., "STransFuse: Fusing Swin transformer and convolutional neural network for remote sensing image semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10990–11003, 2021.
[47] H. Bao, L. Dong, and F. Wei, "BEiT: BERT pre-training of image transformers," 2021, *arXiv:2106.08254*.
[48] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.
[49] H. Zhao et al., "PSANet: Point-wise spatial attention network for scene parsing," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 267–283.
[50] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 418–434.
[51] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.
[52] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[53] H. Wu, C. Liang, M. Liu, and Z. Wen, "Optimized HRNet for image semantic segmentation," *Exp. Syst. Appl.*, vol. 174, Jul. 2021, Art. no. 114532.
[54] S. Guo, Q. Jin, H. Wang, X. Wang, Y. Wang, and S. Xiang, "Learnable gated convolutional neural network for semantic segmentation in remote-sensing images," *Remote Sens.*, vol. 11, no. 16, p. 1922, Aug. 2019.
[55] W. Boulila, "A top-down approach for semantic segmentation of big remote sensing images," *Earth Sci. Informat.*, vol. 12, no. 3, pp. 295–306, Sep. 2019.
[56] J. Zhang, S. Lin, L. Ding, and L. Bruzzone, "Multi-scale context aggregation for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 12, no. 4, p. 701, Feb. 2020.
[57] Y. Sun and W. Zheng, "HRNet- and PSPNet-based multiband semantic segmentation of remote sensing images," *Neural Comput. Appl.*, vol. 35, no. 12, pp. 8667–8675, Sep. 2022.
[58] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 6881–6890.
[59] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7242–7252.
[60] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. NIPS*, 2021, pp. 12077–12090.
[61] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Computer Vision—ECCV*. Glasgow, U.K: Springer, 2020, pp. 213–229.
[62] K. Huang, T. Wu, H. Su, and W. H. Hsu, "MonoDTR: Monocular 3D object detection with depth-aware transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4002–4011.
[63] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.
[64] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. North Amer. Chapter Assoc. Comput. Linguistics*, 2018, pp. 1–16.
[65] A. Ramesh et al., "Zero-shot text-to-image generation," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8821–8831.
[66] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.
[67] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "BiSeNet: Bilateral segmentation network for real-time semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 325–341.
[68] H. Cao et al., "Swin-UNet: UNet-like pure transformer for medical image segmentation," 2021, *arXiv:2105.05537*.
[69] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.



Yuheng Liu received the B.S. degree in electronic and information engineering from Northwestern Polytechnical University, Xi'an, China, in 2021, where he is currently pursuing the master's degree in information and communication engineering with the School of Electronics and Information.

His research interests include remote sensing, image processing, and deep learning.



Yifan Zhang (Member, IEEE) received the B.S. degree in electronics and information technology and the M.S. and Ph.D. degrees in signal and information processing from Northwestern Polytechnical University, Xi'an, China, in 2001, 2004, and 2007, respectively.

From 2007 to 2010, she worked as a Post-Doctoral Researcher with the Vision Laboratory, Department of Physics, University of Antwerp, Antwerp, Belgium. She is currently an Associate Professor with the School of Electronics and Information, Northwestern Polytechnical University. Her research interests include hyperspectral image analysis, image fusion, and image restoration.



Ye Wang received the B.S. degree in automation from the Xi'an University of Science and Technology, Xi'an, China, in 2017, and the M.S. degree in biomedical engineering from Northwestern Polytechnical University, Xi'an, in 2020, where she is currently pursuing the Ph.D. degree in information and communication engineering with the School of Electronics and Information.

Her research interests include remote sensing, hyperspectral object tracking, image processing, and deep learning.



Shaohui Mei (Senior Member, IEEE) received the B.S. degree in electronics and information engineering and the Ph.D. degree in signal and information processing from Northwestern Polytechnical University, Xi'an, China, in 2005 and 2011, respectively.

From October 2007 to October 2008, he was a Visiting Student with The University of Sydney, Camperdown, NSW, Australia. He is currently a Professor with the School of Electronics and Information, Northwestern Polytechnical University. His research interests include hyperspectral remote sensing image processing and applications, intelligent signal and information acquisition and processing, video processing, and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:00:24 UTC from IEEE Xplore. Restrictions apply.